

Deterministic Optimization

Algebraic Aspects of Linear Optimization

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Why Do We Care So Much About
Basic Feasible Solution?

Algebraic Aspects of Linear Optimization

Learning Objectives for this lesson

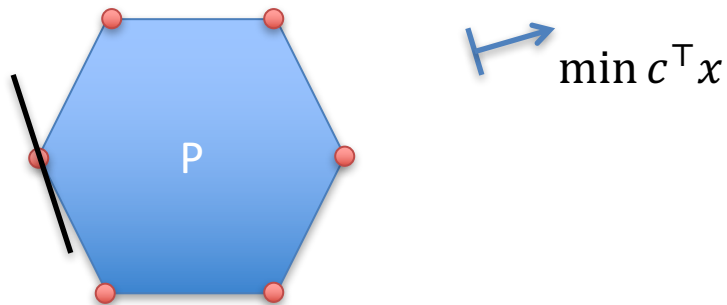
- Identify why basic feasible solution is important for LP
- Recognize what is special about standard form LP regarding basic feasible solution

Which LP has a BFS?

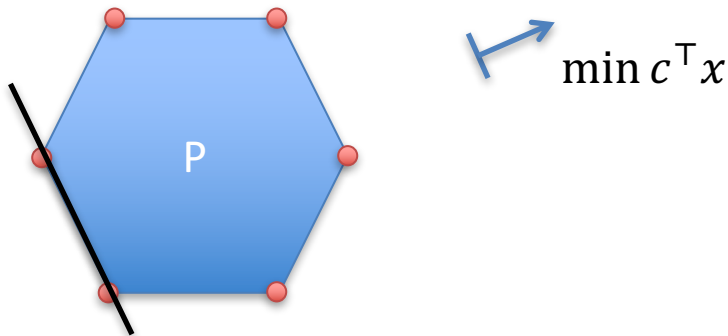
- Not every LP has a BFS, i.e. not every polyhedron has an extreme point
 - Think about a line or a halfspace
- Which LP has a BFS, i.e. which polyhedron has an extreme point?
 - Theorem: A polyhedron P has an extreme point if and only if it does not contain a line.
 - Corollary: **A feasible standard form LP always has a BFS.**
 - Because a nonempty polyhedron in standard form cannot contain a line ($x \geq 0$)

An Optimal Solution is a BFS

- If an LP has a finite optimal solution, then an optimal solution is a BFS.



- This does not mean all optimal solutions must be BFS.



Importance of BFS

- Because feasible standard form LP must have a BFS
- And because an optimal solution must be a BFS
- Then, an optimal solution of standard form LP must be a BFS
- So we only need to look at BFSs, and select the one BFS with the minimum obj cost.
- This is why BFS is very important for linear programming.

Summary

- We learned:
 - Not every polyhedron has an extreme point
 - Not every LP has a BFS
 - But a feasible standard form LP always has at least one BFS
 - If an LP has an optimal solution, then a BFS is optimal